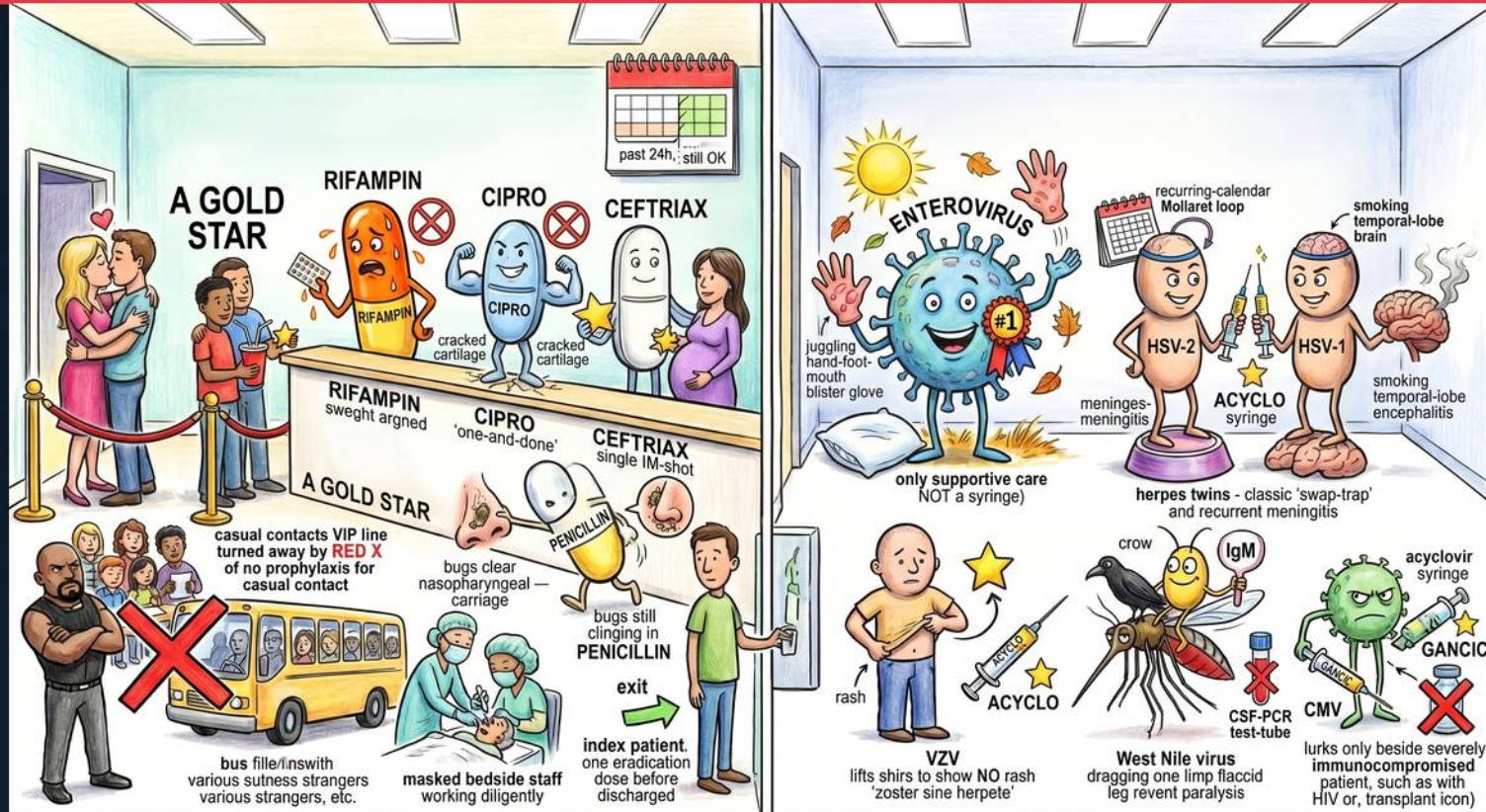
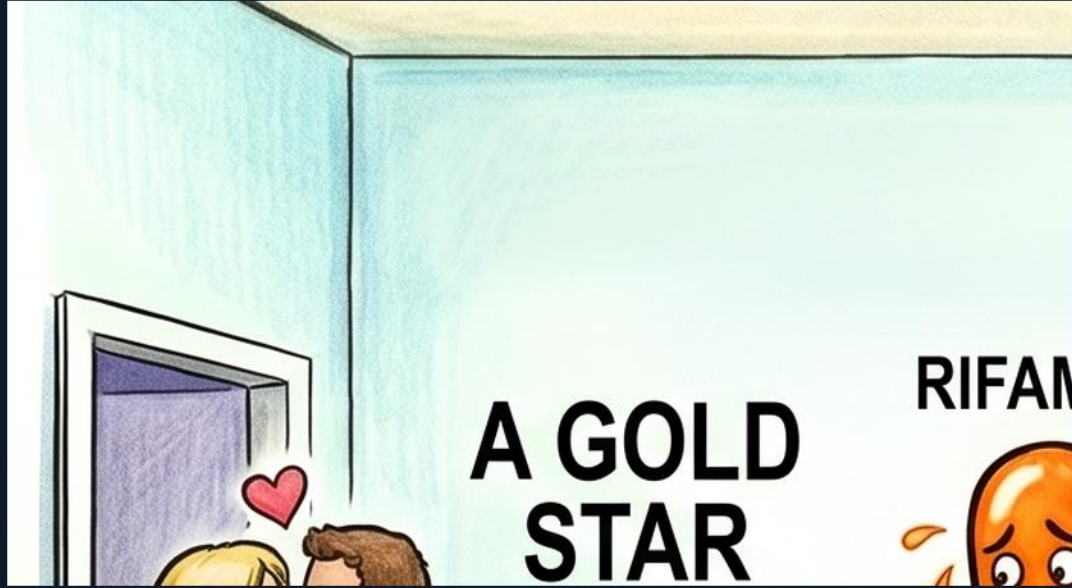


Meningitis — Prophylaxis & Viral Causes





Close contact = gold star (VIP line)

WHAT YOU SEE

The kissing/embracing couple at the velvet rope 'A GOLD STAR' VIP line — intimate close contacts who DO get prophylaxis.

WHAT IT ENCODES

Meningococcal chemoprophylaxis goes ONLY to CLOSE contacts: household members, intimate/kissing contacts, those sharing food/drink/utensils, dorm roommates, barracks-mates, and healthcare workers with unprotected exposure to respiratory secretions (intubation/suctioning without mask) in the 7 days before symptom onset. Goal: eradicate nasopharyngeal carriage and prevent secondary cases.



Casual contacts turned away (red X)

WHAT YOU SEE

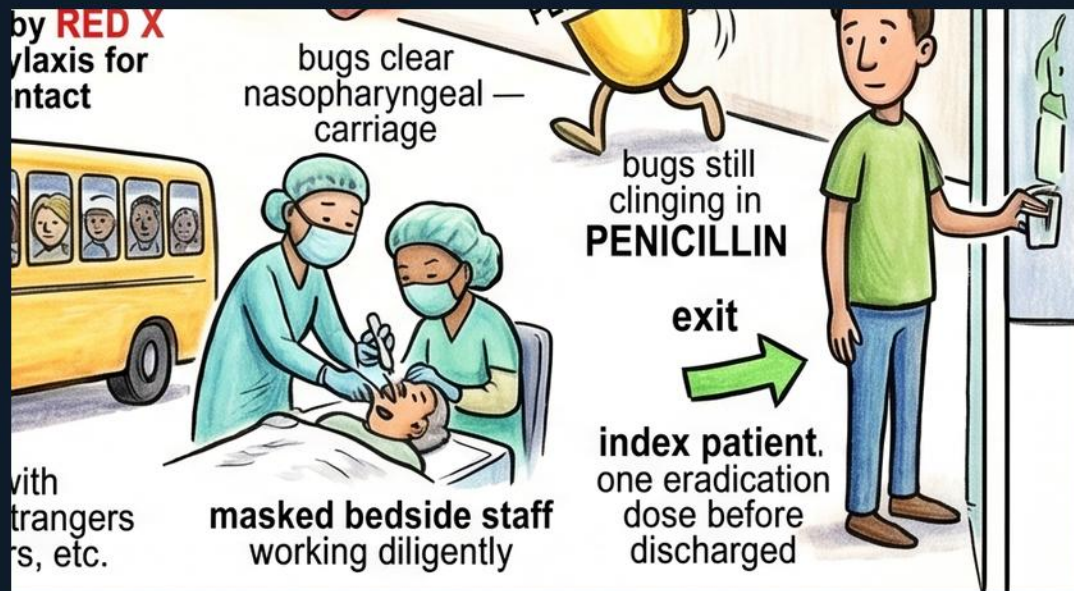
The school-bus crowd of strangers behind a big red X — casual contacts who do NOT qualify for prophylaxis.

WHAT IT ENCODES

Casual contact does NOT warrant prophylaxis: sharing a classroom/office, riding the same bus, or brief bedside care under droplet precautions. Over-prescribing wastes resources and adds resistance/side-effect risk. Eligible agents are rifampin, ciprofloxacin, ceftriaxone (azithromycin as alternative).

THE BOARD TRAP

"give prophylaxis to the whole classroom/office/all hospital staff" -> wrong; casual contact doesn't qualify — restrict to true close contacts within the 7-day window.



Index case eradication on discharge

WHAT YOU SEE

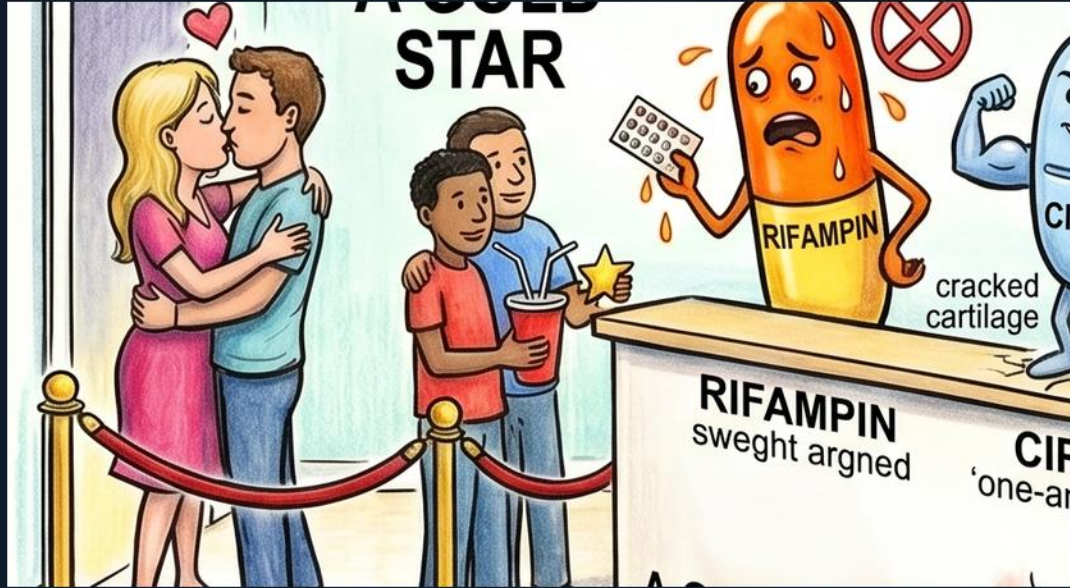
The masked bedside staff / one-pill-before-exit panel — the index patient also needs carriage eradication before going home.

WHAT IT ENCODES

The index patient needs an eradication agent (rifampin, cipro, or ceftriaxone) before discharge **UNLESS** already treated with ceftriaxone/cefotaxime (which clear carriage). IV penicillin or ampicillin treats invasive disease but does **NOT** reliably eliminate nasopharyngeal carriage, so the patient could re-seed contacts.

THE BOARD TRAP

"recovered on IV penicillin, so no longer a carrier" -> wrong; penicillin doesn't eradicate carriage — without a dedicated eradication agent (or a 3rd-gen cephalosporin), the index case can still transmit on discharge.



Rifampin

WHAT YOU SEE

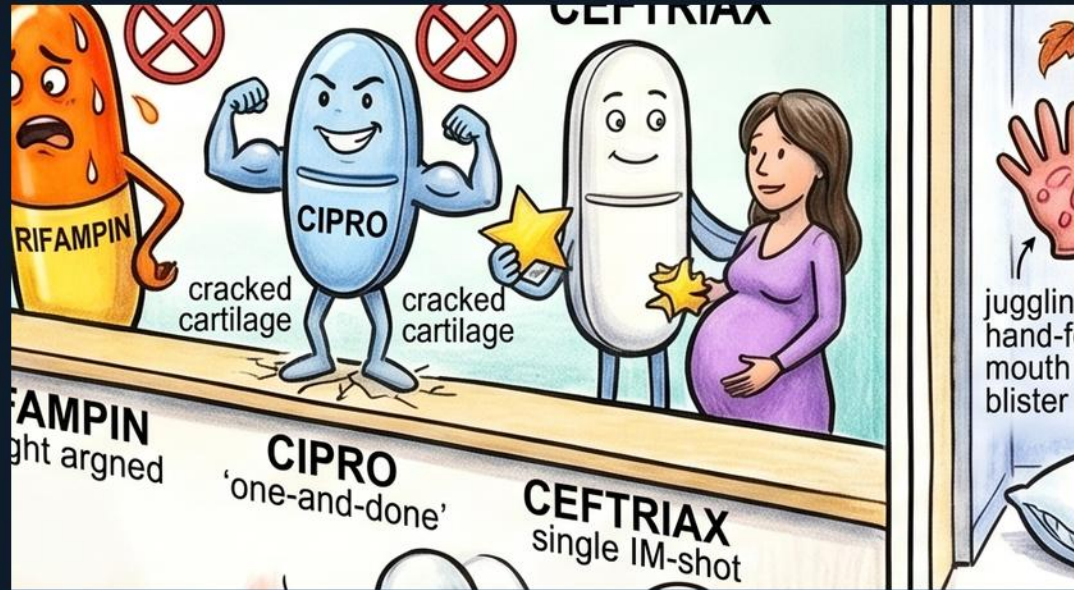
The 'RIFAMPIN' capsule character (with cracked-cartilage red-X warnings nearby) — first-line 2-day prophylaxis.

WHAT IT ENCODES

Rifampin: adults 600mg PO q12h x2 days (4 doses); children $\geq 1\text{mo}$ 10 mg/kg q12h x2d. Counsel about orange-red discoloration of urine/tears/sweat (stains contact lenses) and CYP450 INDUCTION reducing efficacy of oral contraceptives, warfarin, and many drugs.

THE BOARD TRAP

"rifampin for a pregnant contact or someone on OCPs" -> wrong; avoided in pregnancy (use ceftriaxone) and its CYP450 induction can cause OCP failure.



Ciprofloxacin (one-and-done)

WHAT YOU SEE

The flexing 'CIPRO' character labeled 'one-and-done' with cracked-cartilage warning — single-dose adult prophylaxis.

WHAT IT ENCODES

Ciprofloxacin 500mg PO single dose is convenient adult prophylaxis (no multi-dose course); azithromycin is the alternative where fluoroquinolone-resistant meningococcus circulates. Avoid fluoroquinolones in pregnancy and generally in young children (cartilage/tendon concerns).

THE BOARD TRAP

"default to single-dose cipro in a fluoroquinolone-resistant area or for a pregnant/pediatric contact" -> wrong; use azithromycin or ceftriaxone in resistant regions, and ceftriaxone in pregnancy — blanket cipro can fail.



Ceftriaxone (single IM, pregnancy)

WHAT YOU SEE

The 'CEFTRIAX' character labeled 'single IM-shot' — the preferred one-dose option, especially in pregnancy.

WHAT IT ENCODES

Ceftriaxone 250mg IM single dose (125mg for children <15) is the **PREFERRED** prophylaxis in **PREGNANCY** and a reliable single-dose option for anyone. As a 3rd-gen cephalosporin it both treats and eradicates carriage, useful when rifampin or fluoroquinolones are contraindicated.

THE BOARD TRAP

"rifampin or cipro for a pregnant close contact" -> wrong; rifampin and fluoroquinolones are avoided in pregnancy — use single-dose IM ceftriaxone.



Penicillin trap (no carriage eradication)

WHAT YOU SEE

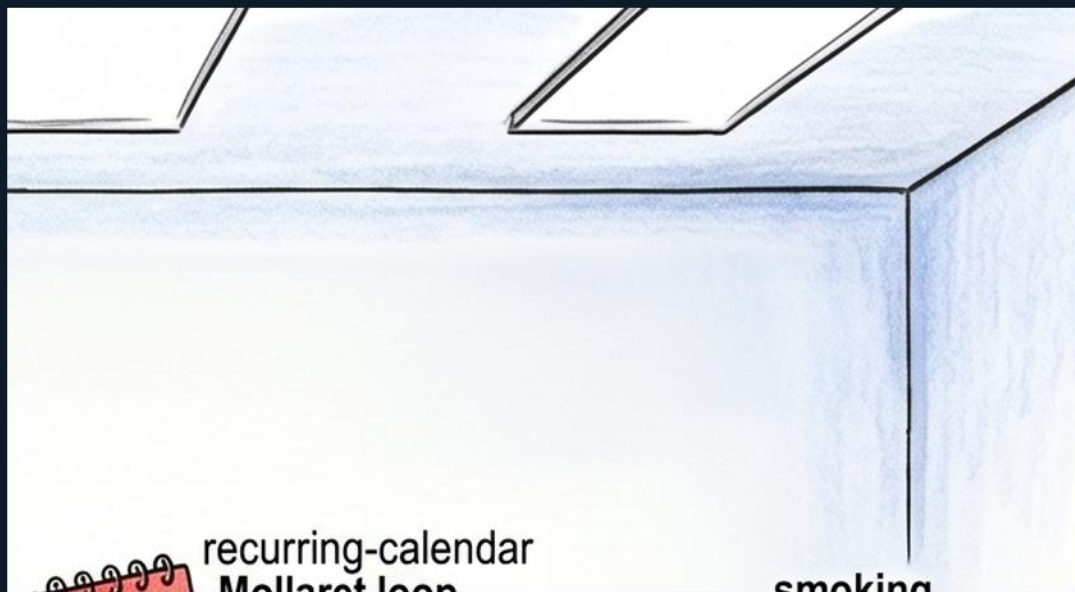
The 'PENICILLIN' nose-character clinging to nasopharyngeal carriage — penicillin treats disease but leaves the bug colonizing.

WHAT IT ENCODES

Penicillin and amoxicillin do NOT reliably eradicate meningococcal nasopharyngeal carriage (poor secretion levels), so they are NOT used for chemoprophylaxis. Acceptable agents: rifampin, ciprofloxacin, ceftriaxone (azithromycin alternative).

THE BOARD TRAP

"oral penicillin or amoxicillin for contact prophylaxis" -> wrong; penicillin treats invasive disease but leaves carriage intact — pick rifampin, cipro, or ceftriaxone.



Prophylaxis timing (past 24h still OK)

WHAT YOU SEE

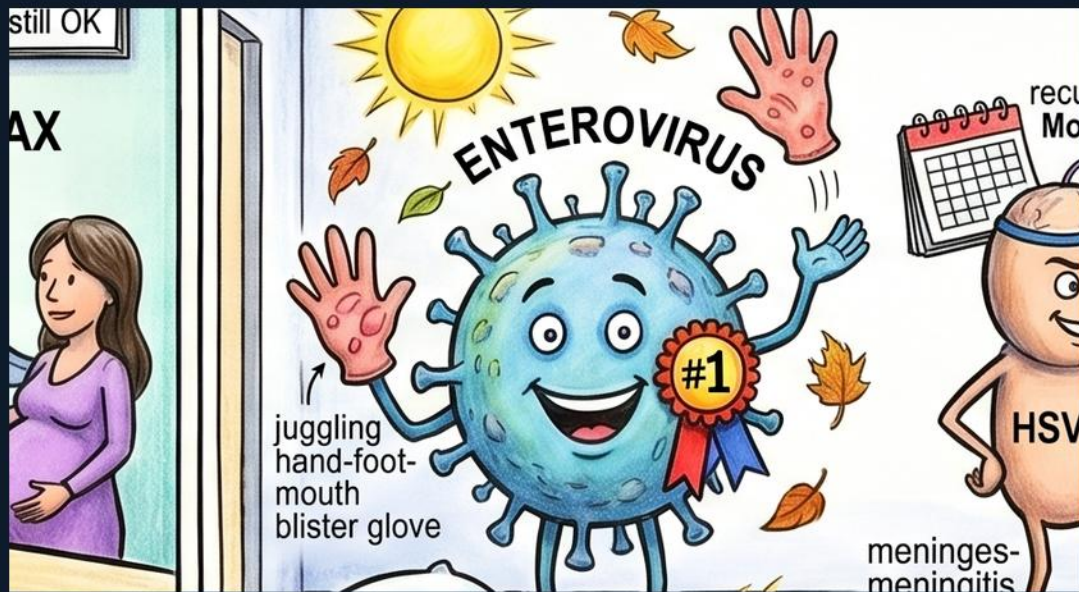
The wall calendar reading 'past 24h... still OK' — the timing myth-buster.

WHAT IT ENCODES

Give prophylaxis ASAP, ideally within 24h of index-case identification, but it remains worthwhile for close contacts up to ~14 days after exposure. The benefit is not confined to the first 24 hours.

THE BOARD TRAP

"decline prophylaxis because >24h has passed" -> wrong; prophylaxis still reduces secondary cases when given within ~2 weeks — withholding it past day 1 misapplies the rule.



Enteroviruses #1 viral cause

WHAT YOU SEE

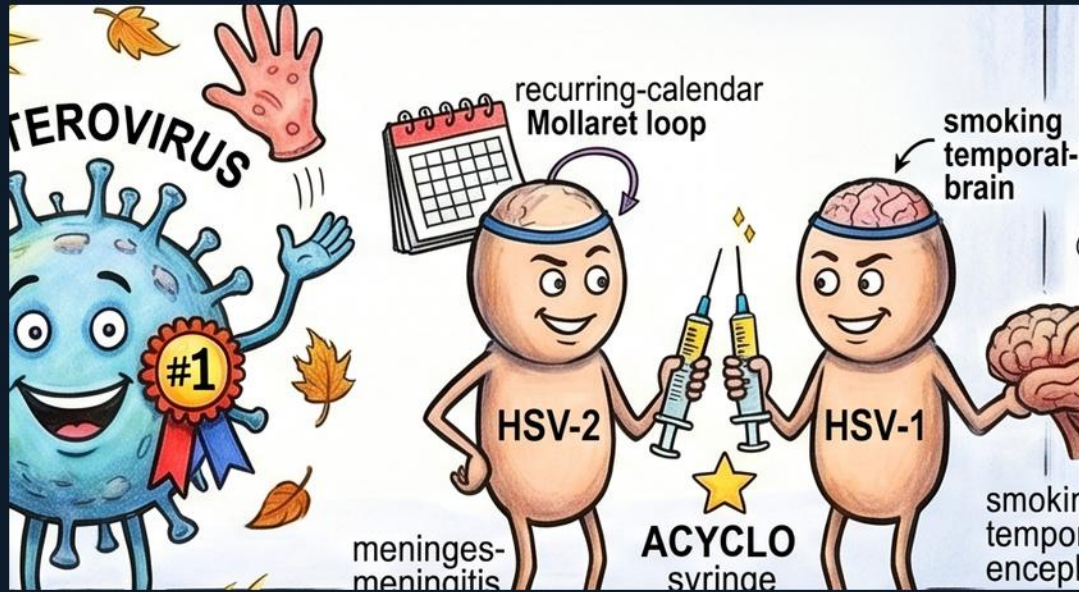
The cheerful spiky blue virus juggling a hand-foot-mouth blister glove, badge '#1' — enteroviruses, the top viral cause.

WHAT IT ENCODES

Enteroviruses (coxsackie, echovirus) cause >85% of viral (aseptic) meningitis, peaking SUMMER/FALL via fecal-oral spread, sometimes with hand-foot-mouth, herpangina, or myopericarditis. CSF is lymphocytic with normal glucose; diagnose by CSF RT-PCR. Management is SUPPORTIVE — no specific antiviral.

THE BOARD TRAP

"start antibiotics or acyclovir for a well patient with summertime lymphocytic/normal-glucose CSF and enterovirus PCR" -> wrong; enteroviral meningitis is self-limited — supportive care suffices (empiric antibiotics reasonable only until bacterial excluded).



HSV-2 = meningitis (Mollaret)

WHAT YOU SEE

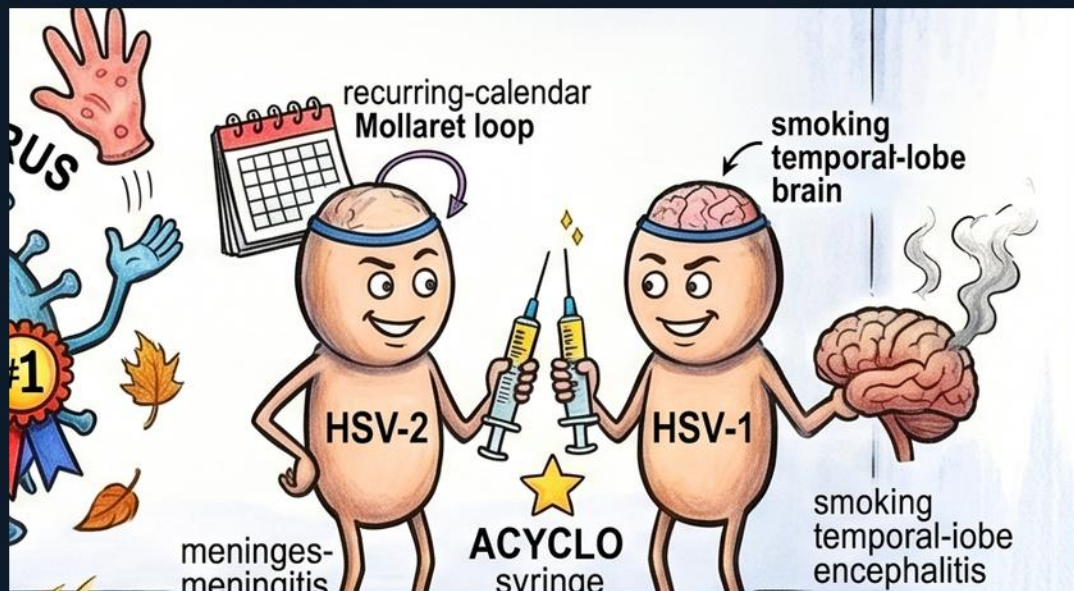
The left 'herpes twin' with the recurring-calendar Mollaret-loop and ACYCLO syringe — HSV-2 causing recurrent aseptic meningitis.

WHAT IT ENCODES

HSV-2 typically causes (often recurrent) aseptic MENINGITIS — recurrent lymphocytic 'Mollaret' meningitis is the classic association, frequently with genital herpes. Diagnose by CSF HSV PCR; treat with IV acyclovir.

THE BOARD TRAP

"HSV-1 causes meningitis, HSV-2 causes encephalitis" -> wrong; the boards test the reverse — HSV-2 -> recurrent meningitis (Mollaret), HSV-1 -> temporal-lobe encephalitis.



HSV-1 = temporal-lobe encephalitis

WHAT YOU SEE

The right 'herpes twin' with the smoking temporal-lobe brain — HSV-1 causing hemorrhagic temporal-lobe encephalitis.

WHAT IT ENCODES

HSV-1 typically causes ENCEPHALITIS with temporal-lobe involvement (hemorrhagic; MRI/EEG temporal changes; RBCs in CSF). Treat with IV acyclovir; start empirically when suspected because delay risks irreversible damage.

THE BOARD TRAP

"wait for HSV PCR before treating suspected HSV-1 encephalitis" -> wrong; start empiric acyclovir immediately — waiting risks irreversible temporal-lobe damage.

VZV — zoster sine herpette

WHAT YOU SEE

The figure lifting a shirt to show **NO** rash beside the ACYCLO syringe — VZV CNS disease without any skin lesions.

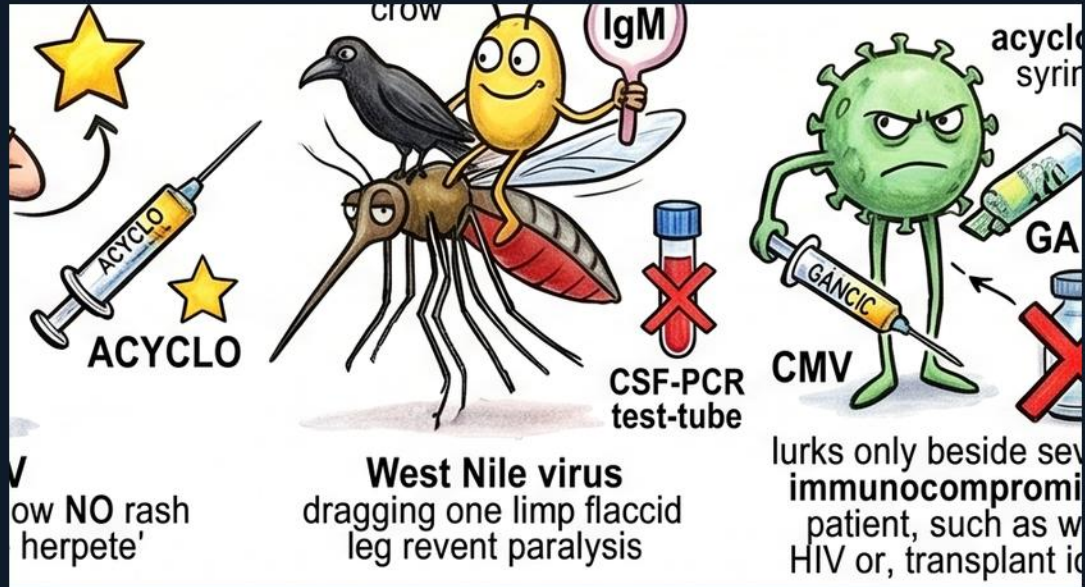
WHAT IT ENCODES

VZV can cause meningitis, encephalitis, vasculitis/stroke, and myelitis, and may occur **WITHOUT** rash (zoster sine herpette), especially in older/immunocompromised patients. Diagnose by CSF VZV PCR and/or CSF VZV IgM; treat with IV acyclovir.

THE BOARD TRAP

"no vesicular rash, so exclude VZV CNS disease" -> wrong; zoster sine herpette means VZV meningitis/encephalitis can present with no rash — send CSF VZV PCR.





West Nile — CSF IgM, flaccid paralysis

WHAT YOU SEE

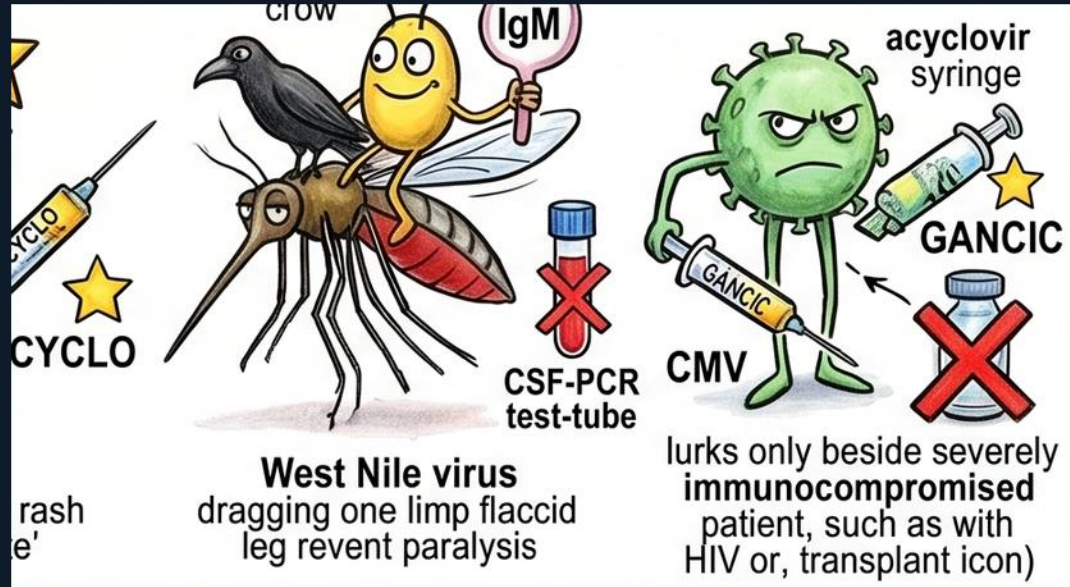
The mosquito/crow dragging a limp leg with the 'IgM / CSF-PCR test-tube' icons — West Nile virus, diagnosed by IgM, causing flaccid paralysis.

WHAT IT ENCODES

West Nile is mosquito-borne (*Culex*) with birds as reservoir, peaks late summer, hits OLDER adults, and classically causes acute asymmetric FLACCID paralysis (anterior horn, polio-like). CSF is lymphocytic with normal glucose; diagnose by CSF (and serum) IgM.

THE BOARD TRAP

"order CSF viral PCR alone to diagnose West Nile" -> wrong; WNV PCR has LOW CSF sensitivity (viremia clears before neuroinvasion) — CSF IgM antibody is the test of choice.



CMV in immunocompromised — ganciclovir

WHAT YOU SEE

The lurking green virus by the bedside with the 'GANCIC' syringe — CMV CNS disease, a disease of severe immunosuppression treated with ganciclovir.

WHAT IT ENCODES

CMV meningoencephalitis (also ventriculoencephalitis, polyradiculopathy, retinitis) occurs almost exclusively in the IMMUNOSUPPRESSED — advanced HIV (CD4 <50), transplant recipients. Diagnose by CSF CMV PCR; treat with ganciclovir (often + foscarnet) plus immune reconstitution. Acyclovir and ganciclovir are NOT interchangeable: acyclovir for HSV/VZV, ganciclovir for CMV (acyclovir fails because CMV lacks thymidine kinase).

THE BOARD TRAP

"acyclovir for CMV (or CMV high on the differential in a healthy host)" -> wrong; acyclovir has no meaningful CMV activity (CMV lacks thymidine kinase), and CMV CNS disease is essentially limited to severe immunosuppression — in a healthy host think enterovirus or HSV-2.

Summary — every symbol

1. Close contact = gold star (VIP line)

Meningococcal chemoprophylaxis goes ONLY to CLOSE contacts: household members, intimate/kissing contacts, those sharing food/drink/utensils, dorm roommates, barracks-mates, and healthcare workers with unprotected exposure to respiratory secretions (intubation/suctioning without mask) in the 7 days before

3. Index case eradication on discharge

The index patient needs an eradication agent (rifampin, cipro, or ceftriaxone) before discharge UNLESS already treated with ceftriaxone/cefotaxime (which clear carriage)

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7. Penicillin trap (no carriage eradication)

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11. HSV-1 = temporal-lobe encephalitis

HSV-1 typically causes ENCEPHALITIS with temporal-lobe involvement (hemorrhagic; MRI/EEG temporal changes; RBCs in CSF)

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2. Casual contacts turned away (red X)

Casual contact does NOT warrant prophylaxis: sharing a classroom/office, riding the same bus, or brief bedside care under droplet precautions

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Rifampin: adults 600mg PO q12h x2 days (4 doses); children $\geq 1\text{mo}$ 10 mg/kg q12h x2d

6. Ceftriaxone (single IM, pregnancy)

Ceftriaxone 250mg IM single dose (125mg for children <15) is the PREFERRED prophylaxis in PREGNANCY and a reliable single-dose option for anyone

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